

# Workshop 1: Accessing Command Line Interfaces

CSCI3251 / AIST2602  
Engineering Practicum

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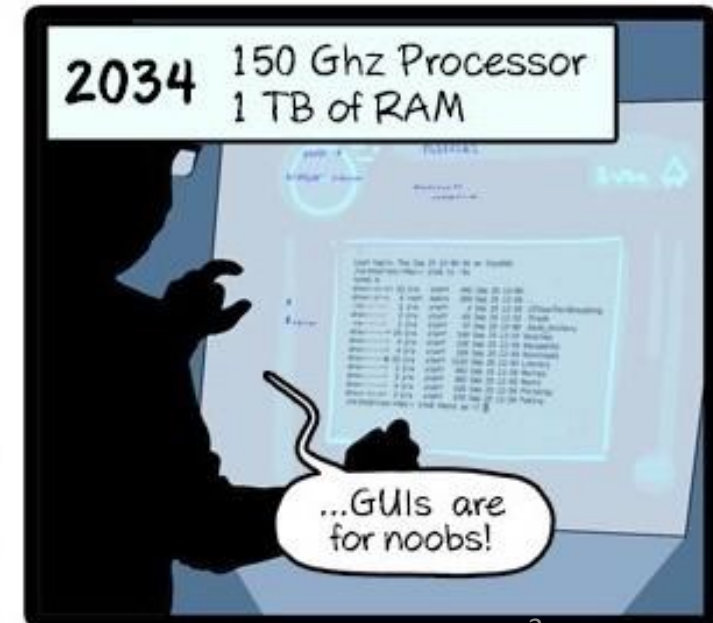
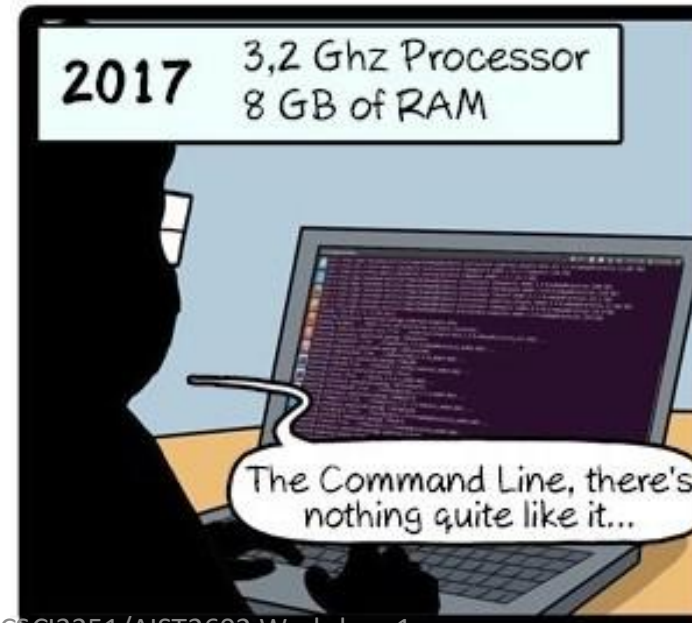
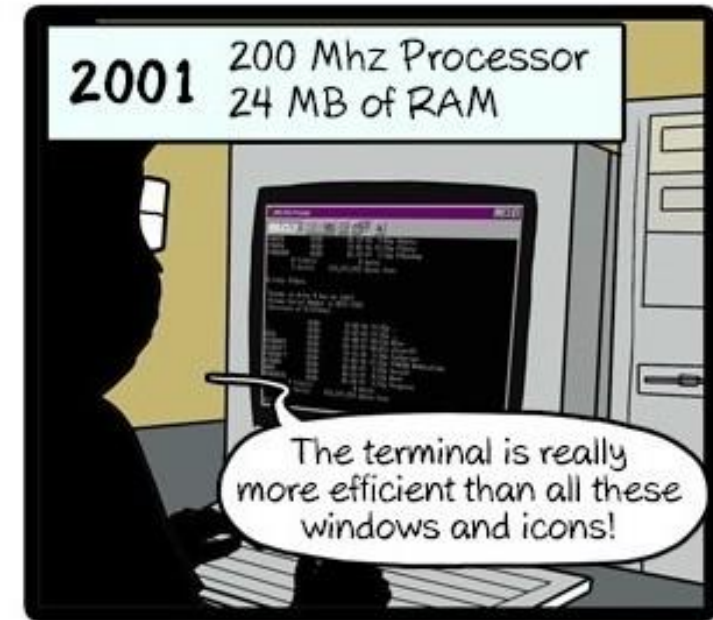
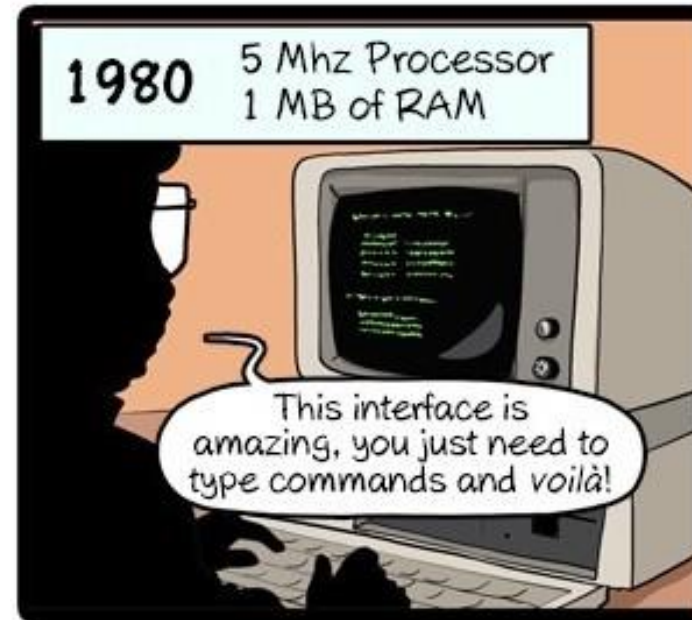
# Workshop 1:

## Accessing Command Line Interfaces

- The CLI
- What is a shell?
- SSH
- CSE servers

# The CLI

- Have you been using it?
- Why do you use it?
- Do you like it?



# The CLI

- Why command line?
- Examples...
  - File system tasks
  - Data processing using scripts
  - Code compilation
  - Remote server management
- Windows: Command Prompt / Powershell
- macOS: Terminal

# The Linux operation system

- Widely tested for stability in networking and server environments
- Slightly more developed on CLI, yet GUI is still available
- The Linux OS
  - Kernel
    - The “official” core of Linux
  - Distributions
    - Ubuntu, CentOS, RedHat, ...
  - Shells
    - bash, csh, zsh, ...

# Different ways to access Linux

- Installing Linux onto your computer
  - As the primary OS, a dual-boot OS, or a virtual machine
- Cloud services, e.g., AWS, Azure, etc.
  - Easily accessible with SSH or even web platforms
  - Maybe costly, depending on computation needs and usage
  - Free-tier available
- CSE Linux servers
  - No administrative rights
  - Tasks without intense computing resources can be done

# Different ways to access Linux

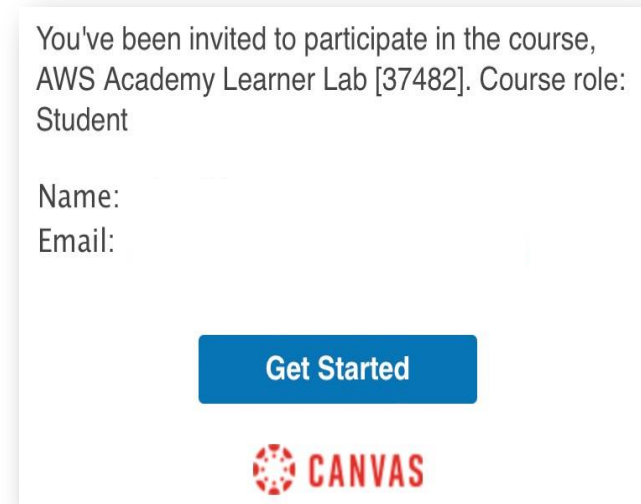
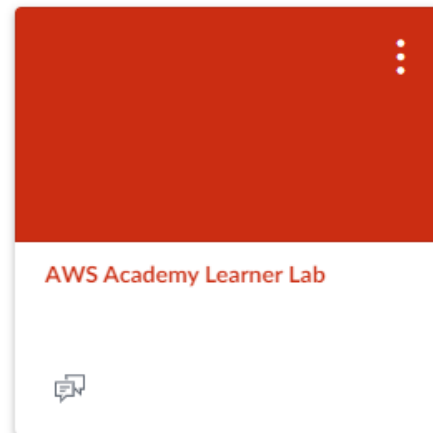
- ***Windows Subsystem for Linux*** (WSL)
  - A subset of the Linux system (kernel and shell) being an “app” on Windows
- Install it with instructions from Microsoft
  - <https://docs.microsoft.com/en-us/windows/wsl/about>
- Good enough for basic tasks!

# Different ways to access Linux

- ***Terminal*** on Mac
- Mac is UNIX based
  - \*nix refers to Linux, UNIX, and similar OSes
- Most commands in Linux are based on the original UNIX version
  - Subtle differences nowadays when two communities develop separately

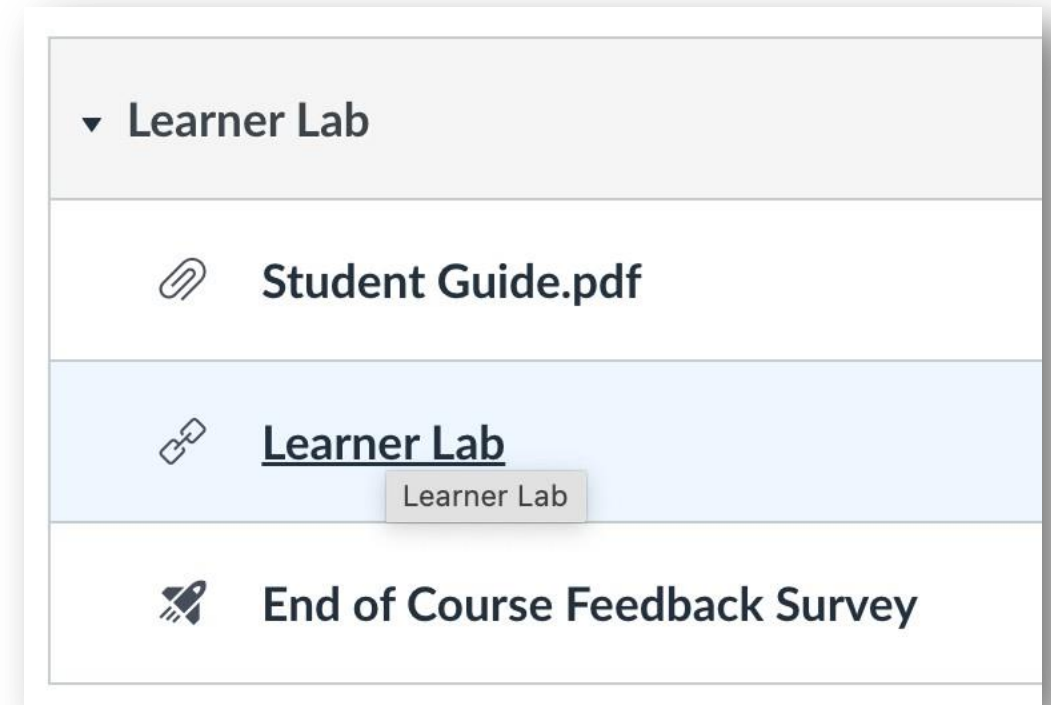
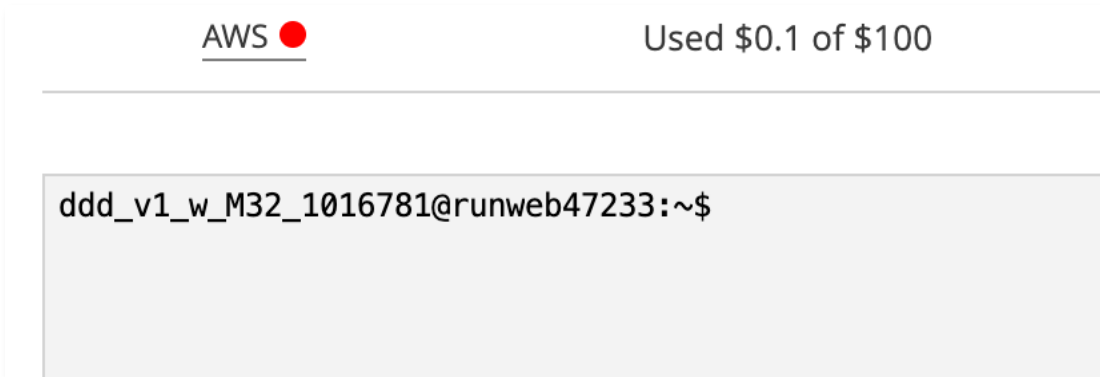
# Getting into your AWS Academy account

- You should have received an email at your @link email for AWS
  - Sign in at <https://awsacademy.instructure.com/login/canvas>
  - or [https://www.awsacademy.com/vforcesite/LMS\\_Login](https://www.awsacademy.com/vforcesite/LMS_Login)
  - Click student login, and
- Choose the button and activate your account
- In the ***Dashboard***, choose the ***AWS Academy Learner Lab***



# Getting into your AWS Academy account

- Under Modules, choose Learner Lab – Foundational Services
- It takes a while to setup
- Then, you should arrive at a shell!



# Commands

- Most of \*nix commands follow this basic structure

**command** *[option(s)] [argument(s)]*

- *Whenever you see [ ] in the documentations, that part is optional*
- The **command** is usually an executable in /bin, /sbin, /usr/bin, or other binary directories
  - Try the **whereis** command to find where an executable is

# Commands to know about the machine

- **uname -a**
  - The version of Linux, date, CPU architecture, etc
- **lscpu**
  - Information of the CPU
- **top**
  - Current running processes from various users
  - *(Press **q** to quit)*
- **free -h**
  - Memory usage

# Accessing the Linux filesystem

- The Linux filesystem starts at /
  - The “root”
- Everything in the computer is under root
  - E.g., user files, system files, mounted harddisks, etc.
- Some abbreviations:
  - Your home directory: ~
  - Your current directory: .
  - The parent directory: ..
- System directories might include...
  - **/bin**: Essential programs
  - **/dev**: Device files
  - **/etc**: Config files
  - **/home**: User homes
  - **/lib**: Essential shared libraries
  - **/proc**: Kernel and processes
  - **/tmp**: Temporary files
  - **/usr**: User applications
  - **/var**: Data files (e.g., system log)

# Important commands

- *You may consider “**directory**” as “**folders**” which you have seen in other OSes*
- **id**
  - Information of user
- **pwd**
  - What’s the current directory?
- **ls**
  - List files at current directory
  - **-a**: show all files
  - **-l**: show long format
- **cd**
  - Change to another directory
- **cp**
  - Copy files
- **mv**
  - Move files (or rename files!)
- **rm**
  - Delete files
- **mkdir**
  - Create a new directory

# Documentations

- There are too many useful commands you can ever learn
- Offline documentations:
  - Manual pages (*press **q** to quit*)

**man** *[command]*

- Online documentations:
  - English: <https://www.linux.org/forums/linux-beginner-tutorials.123/>
  - Chinese: <https://linux.vbird.org>
  - There are too many!

# SSH

- Some Linux computers are provided for remote access at the CSE department
  - You need your CSE account for access
- No administrative rights
- Tasks without intense computing resources can be done
  - e.g., batch processing of simple data
- See: <https://i.cse.cuhk.edu.hk/technical/linux-service/cluster/>
  - *(Only accessible via CUHK/CSE VPN)*

# What is your CSE account?

- You are entitled for access to different systems in the CSE department
  - <https://i.cse.cuhk.edu.hk/technical/>
- You were given a CSE account some time after enrolling into CS/CE/AIST



DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING  
THE CHINESE UNIVERSITY OF HONG KONG  
USER ACCOUNT MEMO

NAME : XXX XXX XXX  
DATE : 31 Aug 2020

Dear XXX XXX XXX,

Your account has been created with following information:

|                |                     |                  |
|----------------|---------------------|------------------|
| SYSTEMS        | : Linux             | (linuxXX)        |
| USERNAME       | : xxxxxxx0          |                  |
| PASSWORD       | : -----             | [sample: 1l, 00] |
| DISK QUOTA     | : 2.5GB             |                  |
| HOME DIRECTORY | : /uac/y20/xxxxxxx0 |                  |

-----

|          |              |
|----------|--------------|
| SYSTEMS  | : Windows NT |
| USERNAME | : xxxxxxx0   |
| PASSWORD | : -----      |

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Please change your password immediately after the first login. On Linux, login any Linux machines (e.g. linux10) and use command "yppasswd" On Windows, press <Ctrl><Alt><Del> and then select <Change Password>.

DON'T DISCLOSE YOUR PASSWORD TO ANY OTHER PERSON.

For any question in using the computing systems, please check the "Frequently Asked Questions" in CSE Intranet:  
<https://i.cse.cuhk.edu.hk/technical/>

OR contact System Administrator by either email to techteam@cse.cuhk.edu.hk, phone at 3943 8999 or approaching technical team counter in Room 1001.

System Administrator

Email [techteam@cse.cuhk.edu.hk](mailto:techteam@cse.cuhk.edu.hk) if you have issues...

# SSH to CSE servers

- You can connect to CSE Linux servers from anywhere

ssh [username@gw.cse.cuhk.edu.hk](mailto:username@gw.cse.cuhk.edu.hk)

```
eee_W_2748332@runweb109902:~$ ssh umair@gw.cse.cuhk.edu.hk
umair@gw.cse.cuhk.edu.hk's password:
Linux nile 3.2.0-4-amd64 #1 SMP Debian 3.2.60-1+deb7u3 x86_64
```

- gw.cse is a login gateway
- Then you can connect to the linux server you want

```
<myHome> ssh linux1
Warning: Permanently added the RSA host key for IP address '137.189.88.14' to the list of known hosts.
umair@linux1's password:
Welcome to Ubuntu 18.04.6 LTS (GNU/Linux 4.15.0-171-generic x86_64)
```

- You might need to answer “Yes” in your first connection
- If you are in the CUHK network, you can directly ssh to *linux?.cse.cuhk.edu.hk* in the first place

# SSH to CSE servers

- Same ssh command if you have macOS Terminal, or WSL
- There are lots of GUI SSH clients you can use for your own OS
  - e.g., MobaXterm, PuTTY
- Some includes file transfer functionalities
- If you need further help with connection to CSE servers, see <https://i.cse.cuhk.edu.hk/technical/internet-wifi-vpn-ssh-web-hosting/>

# Exercise

- Type these commands in any CSE server
  - **w**
  - **id**
  - **uname -a**
- Make a screenshot of the results, and submit the image to Blackboard
- If you have any issues with your CSE account, please email [techteam@cse.cuhk.edu.hk](mailto:techteam@cse.cuhk.edu.hk) with your @link email, and state your name and SID clearly

# Summary

- You have learnt:
  - Accessing AWS shell
  - Some basic commands
  - Accessing CSE servers
- Submit to Blackboard for Workshop 1
  1. Screenshot of running some commands
  - Submission deadline: **Monday - 20/Jan/2025, Time: 23:59**